Wonguk Lee

Ph.D. Student, Management Information Systems (MIS)
Business and Technology Management (BTM), KAIST
Advisor: Prof. Chul Ho Lee

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1. EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)**
  Ph.D. Student in Business and Technology Management (BTM)
  Major: Management Information Systems (MIS)
- **Korea Advanced Institute of Science and Technology (KAIST)**
  M.S. in Business and Technology Management (BTM)
  Major: Management Information Systems (MIS)
- **Handong Global University (HGU)**
  B.B.A. in Business Administration
  B.A. in Law (Double Major)

2. RESEARCH INTERESTS

- Conversational AI and mental healthcare
- Mobility and geographic access to essential services
- Digital platforms and AI governance
- Technological innovation and knowledge flows

3. PROJECTS

- **Onto-LLM-Based Global Outbound Recruitment Service Development**
  TIPS R&D Program, Ministry of SMEs and Startups (MSS) | Aug 2025 – Jul 2027 | Ongoing
  Contribute to ontology-based recruitment AI, Graph-RAG / Graph-of-Thought reasoning, and agent prototype research.
- **Open-Web-Based Automated Resume Collection and AI Recruitment Support System**
  Startup Growth Technology Development Program, TIPA / MSS | Aug 2024 – Jul 2025 | Completed
  AI recruitment and data-pipeline development.
- **AI Counseling Chatbot for Schizophrenia Care**
  Jeonbuk National University Hospital | Ongoing
  Develop and study a free patient-facing LLM/RAG chatbot, focusing on perceived judgment, stigma-related engagement barriers, disclosure, and supportive interaction design.
- **Smart Mobility Safety Services and Infrastructure for Autonomous Driving**
  MOTIE / KIAT, International Industrial Technology Cooperation | Project P0025238
  2025 | Technical development and patent contribution
  Developed OSM-supported road-detection and road-geometry algorithms and physics/XGBoost-based speed recommendations with KAIST and industry collaborators.
- **Public Bike-Sharing and Transit Integration for Local Mobility and Consumer Activity**
  Yuseong District Office and Daejeon Transportation Corporation support | Project lead
  Secured and led the project; integrated Tashu bike-sharing, transit, Shinhan Card spending, and KT population-movement data to study first/last-mile connections, local festivals, and economic activity.
- **Personal Mobility Device Adoption among Users and Non-users**
  Behavioral research | Manuscript under review
  PLS-SEM and multi-group analysis of 444 respondents, including 222 users and 222 non-users; examined perceived physical risk, trust, and adoption intentions.

- **Multiple-RAG and Multidimensional Ontology-Based Recruitment sLLM**

  Didimdol R&D planning | 2026
  Research and engineering design for a recruitment-specific small language model, evidence retrieval, ontology mapping, evaluation, and an end-to-end processing pipeline.

- **SF-Olive Smart Farm Platform for Olive Production**

  International Collaborative Technology Development | Planned project period Aug 2026 – Jul 2029
  KAIST participating researcher role: data experimentation and bias-analysis support for a platform combining advanced sensing, blockchain, and edge AI semiconductors.

4. PUBLICATIONS AND WRITING

- **Published and forthcoming**

(1) ***Sectoral Heterogeneity in the Digital Transformation-Financial Performance Nexus: Evidence from Korean Manufacturing Firms***

Junehwan Kim†, **Wonguk Lee**†, and Euiseok Kim
† Equal contribution.
Accepted · Forthcoming in Asian Journal of Technology Innovation

(2) ***Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts***

Wonguk Lee, Donggyu Min, Antino Kim, and Chul Ho Lee
ICIS 2026 Proceedings, Healthcare and Wellbeing, Paper 19. Completed research paper.
Proceedings

(3) ***Transformation of Business Models in Metaverse Era: Through a Systematic Literature Review and Bibliometric Analysis***

Temitayo Shenkoya, **Wonguk Lee**, and Euiseok Kim (2024).
Journal of Information Technology Services, 23(6), 1–17.
DOI: 10.9716/KITS.2024.23.6.001

(4) ***A Study on the Flow of Technological Knowledge in Korean Industries: Focusing on Inter-industry Patent Citation Analysis Based on Pavitt's Taxonomy***

Wonguk Lee and Euiseok Kim (2024).
Journal of Korea Technology Innovation Society, 27(3), 431–454.
DOI: 10.35978/jktis.2024.6.27.3.431

(5) ***A Study on the Intention to use Personal Mobility Services: Focused on the SOR (Stimulus-Organism-Response) Model***

Wonguk Lee and Heetae Yang (2022).
Information Systems Review, 24(2), 67–88.
DOI: 10.14329/isr.2022.24.2.067

- **Working papers**

(6) ***Formation and Dissipation of the Hypernetwork Effect: Transitional Complementarity between Learning and Participation in AI-Driven Networks***

Wonguk Lee and Euiseok Kim
Under revision · Second round

(7) ***Primary Token Issuance and Liquidity-Segmented Returns in Crypto Markets***

Kyeong Hun Kim, **Wonguk Lee**, and Byung Hwa Lim
Under revision · Minor revision

(8) ***Understanding corporate digital entrepreneurship in financial institutions: The roles of AI adoption and entrepreneurial orientation***

Heetae Yang and **Wonguk Lee**
Submitted

(9) ***Understanding Personal Mobility Device Adoption Through the S–O–R Framework: A Multi-Group Analysis of Users and Non-users***

Wonguk Lee and Heetae Yang
Under review

(10) ***The Effects of Public Bike-Sharing (Tashu) and Public Transit Integration on Local Festivals and Consumer Spending***

Kyusung Kim, **Wonguk Lee**, Haeun Jang, and Hyungu Jang
Under revision · Excellent Paper Award, Korea Society of IT Services

(11) ***The Role of NFTs in the Digital World: Catalyzing Innovation in the Creative Industry through a Patent-Inspired Approach***

Wonguk Lee and Euiseok Kim
Under review

- **Work in progress**

(12) ***Breadth to Enter, Relatedness to Stay: Technology Portfolios and the Limits of Firms' Green Diversification***

Wonguk Lee and Euiseok Kim
Work in progress

(13) ***Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts***

Wonguk Lee, Donggyu Min, Antino Kim, and Chul Ho Lee
Work in progress; related ICIS 2026 proceedings version listed above.

(14) ***Protecting the Investor, Punishing the Retailer: The Perverse Effects of Banning AI Autonomous Investment***

Wonguk Lee, Gunsu Kim, and Chulho Lee
Work in progress · Accepted for presentation at WITS; presentation forthcoming.

(15) ***Funded but Not Followed? Donor Relationships After Sponsor-Led Funding Completion***

Wonguk Lee and Chul Ho Lee
Work in progress

(16) ***AI Summary Design in Consumer Complaint Platforms***

Wonguk Lee and Chul Ho Lee
Work in progress · Working title

- **Books and chapters**

- **Generative AI as a Creative Collaborator**

  Lee, W. (2026). In AI Transformation Era: Innovation and Challenges of Korean Industries Highlighted by KAIST. Yulgok Publishing.

- **AI Era: Structure and Sustainability of the Korean Workforce Ecosystem**

  Co-author. Selected as an Excellent Academic Book by the National Academy of Sciences, Republic of Korea (2025).

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5. PATENTS

- **Ontology-Based Data Standardization for Human Resource Management**

  Registered. Korean patent 10-2891011-00-00 · Registered 20 November 2025. Korean application 10-2025-0100830 · Filed 24 July 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

- **Ontology-Based Human Resource Analysis and Job Inference**

  Registered. Korean patent 10-3027368-00-00 · Registered 29 September 2026. Korean application 10-2025-0174900 · Filed 18 November 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

- **LLM-Based Ontology Data Processing for Human Resource Information**

  Registered. Korean patent 10-3027373-00-00 · Registered 29 September 2026. Korean application 10-2025-0176365 · Filed 19 November 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

- **Apparatus and Method for Ontology-Based Candidate Search**

  Registered. Korean patent 10-3035190-00-00 · Registered 13 October 2026. Korean application 10-2025-0170399 · Filed 12 November 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

- **Resume Data Augmentation and Personalized Inference Using Relationship-Aware LLM Context**

  Application filed. Korean application 10-2026-0082060 · Filed 7 May 2026. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

- **Talent Search and Recommendation Based on Conversational Hiring-Needs Analysis**

  Application filed. Korean application 10-2026-0110525 · Filed 17 June 2026. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

- **Ontology-Based Data Standardization for Human Resource Management**

  International application filed. International application PCT/KR2026/009822 · Filed 22 July 2026. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee. Related Korean application: 10-2025-0100830.

- **Road Determination Device and Method of Determining a Road**

  Application filed. Korean application 10-2025-0209757 · Filed 24 December 2025

- **Recommended Speed Determination Apparatus and Method**

  Application filed. Korean application 10-2025-0207816 · Filed 23 December 2025

- **Method and Apparatus for Evaluating Image Tags for AI Training**

  Application filed. Korean application 10-2024-0067992 · Filed 24 May 2024

- **Method for Predicting Fracture of Pipe Due to Corrosion of Alloy and System Thereof**

  Application specification. Korean patent application specification · 17 April 2024

6. HONORS AND AWARDS

- **Excellent Paper Award**

  Korea Society of IT Services | Public bike-sharing (Tashu) and public-transit integration study.

- **Outstanding Research Idea Award**

  BTM Seminar, KAIST | When the AI Bot Advisor Leaves: Unpacking Investor Shifts and Loan Imbalances in P2P Lending

- **Excellent Academic Book Selection (National Academy of Sciences, Republic of Korea)**

  AI Era: Structure and Sustainability of the Korean Workforce Ecosystem (Co-author)

7. APPOINTMENTS AND TEACHING

- **Chief Data Officer, TalentSeeker**

  February 2024–present. Full-time C-level role. Lead data ontologies, multi-agent systems, data-pipeline architecture, data engineering, and LLMOps for AI-enabled recruitment.

- **Lead Course Instructor in Data Science, Baekseok University**

  Second semester, 2026. English-medium course design and teaching in Python, pandas, data preparation, visualization, statistics, prediction, and responsible generative AI use.

8. UPCOMING PRESENTATIONS

- **CES 2027**

  Scheduled to attend as a C-level speaker in my role as Chief Data Officer at TalentSeeker.

- **Workshop on Information Technology and Systems**

  Protecting the Investor, Punishing the Retailer: accepted for presentation at WITS; presentation forthcoming.

9. RESEARCH METHODS

Methods: GIS and OSM networks; physics-based modeling and XGBoost; PLS-SEM and multi-group analysis; experiments; panel-data analysis; agent-based simulation; patent analysis. Tools: Python, pandas, data engineering, ontologies, RAG, multi-agent systems, and LLMOps.